H2 ANDERSON JUNIOR COLLEGE HIGHER 2

BIOLOGY 9648/01

Paper 1 Multiple Choice

23 September 2016
Friday

1 hour 15 mins

Additional Materials: Multiple Choice Answer Paper

READ THESE INSTRUCTIONS FIRST

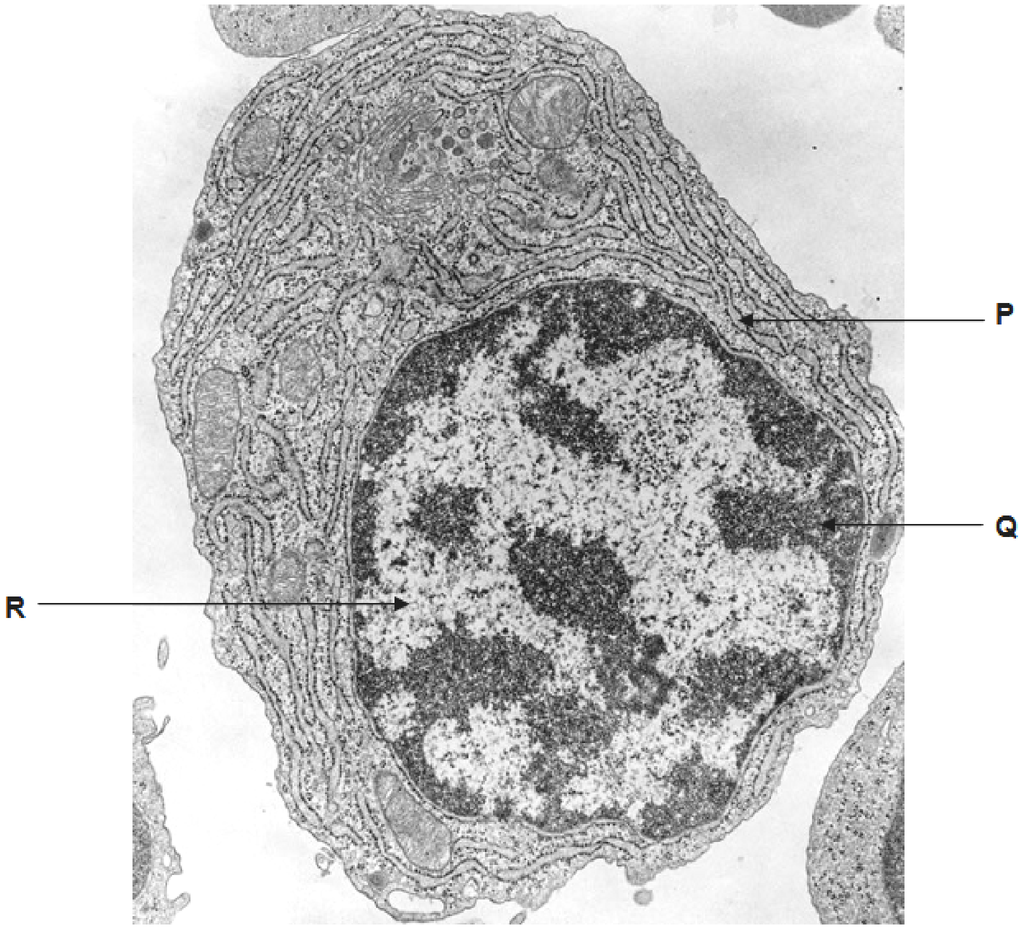
Write in soft pencil.
Do not use staples, paper clips, highlighters, glue or correction fluid.
Write your name, PDG and identification number on the Answer Sheet.

There are **forty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A**, **B**, **C** and **D**.
Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
Any rough working should be done in this booklet.

Calculators may be used.

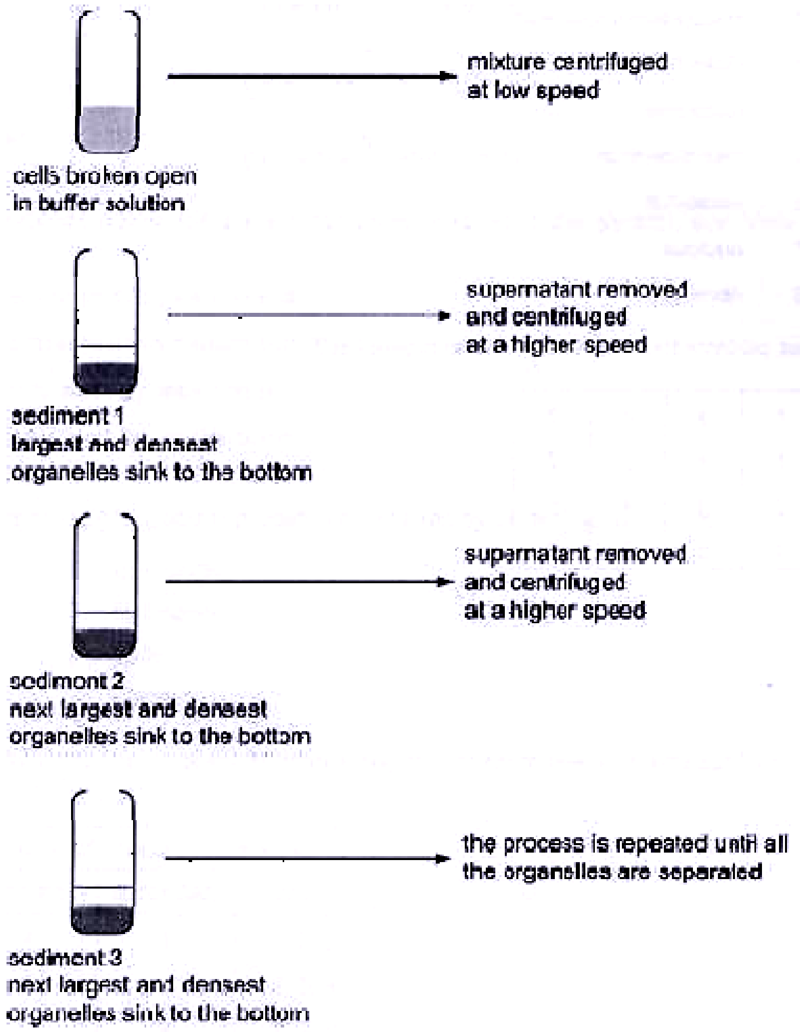
1 The figure below shows an electron micrograph of a cell.



Which of the following about structures **P**, **Q** and **R** are correct?

	P	Q	R
A	provides large surface area for attachment of ribosomes	contains demethylated DNA	contains acetylated histones
B	transport of proteins to Golgi apparatus	histones are deacetylated	active condensation of chromatin
C	synthesis and processing of membrane proteins	contains methylated DNA	active transcription of genes
D	synthesis of phospholipids and steroid hormones	transcription of genes silenced	synthesis of proteins on free ribosomes

2 Fractionation is a process used to separate cell components according to their size and density. The diagram shows the main stages in fractionation of a plant cell.



DCPIP and buffer solution (containing glucose, fructose, sodium bicarbonate) were added to each of the sediments, and the mixtures were left in the dark for fifteen minutes. Sediment 2 caused the DCPIP to be reduced.

Which organelle present in Sediment 2 caused reduction of DCPIP?

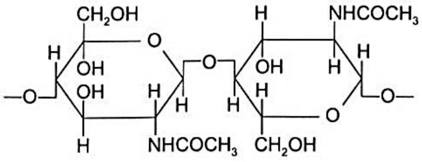
A Chloroplast

B Mitochondria

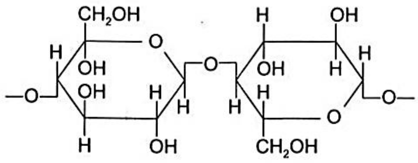
C Nuclei

D Ribosomes

3 The diagrams show short sections of some common polysaccharides and modified polysaccharides.



1



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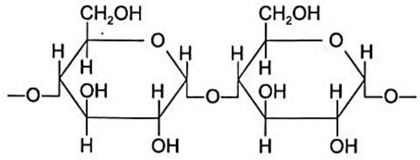

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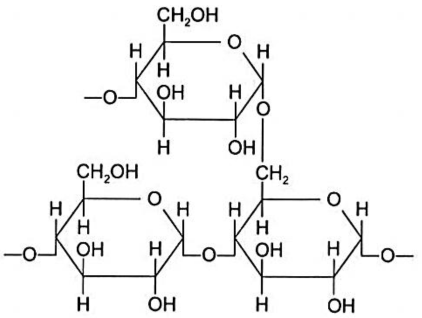

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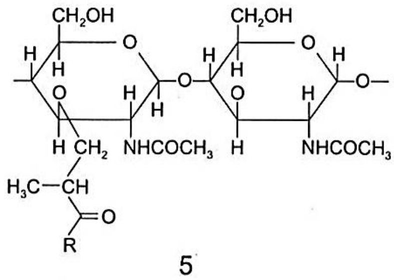


5

The polysaccharides can be described as below.

- polysaccharide **F** is composed of β-glucose monomers with 1,4 glycosidic bonds
- polysaccharide **G** is composed of α-glucose monomers with 1,4 and 1,6 glycosidic bonds
- polysaccharide **H** is composed of N-acetylglucosamine and N-acetylmuramic acid monomers with β-1,4 glycosidic bonds
- polysaccharide **J** is composed of α-glucose monomers with 1,4 glycosidic bonds
- polysaccharide **K** is composed of N-acetylglucosamine monomers with β-1,4 glycosidic bonds

Which shows the correct pairings of polysaccharide descriptions and diagrams?

	polysaccharide **F**	polysaccharide **G**	polysaccharide **H**	polysaccharide **J**	polysaccharide **K**
A	2	4	5	3	1
B	2	5	4	1	3
C	3	4	5	2	1
D	3	5	4	1	2

4 Which description concerning collagen is correct?

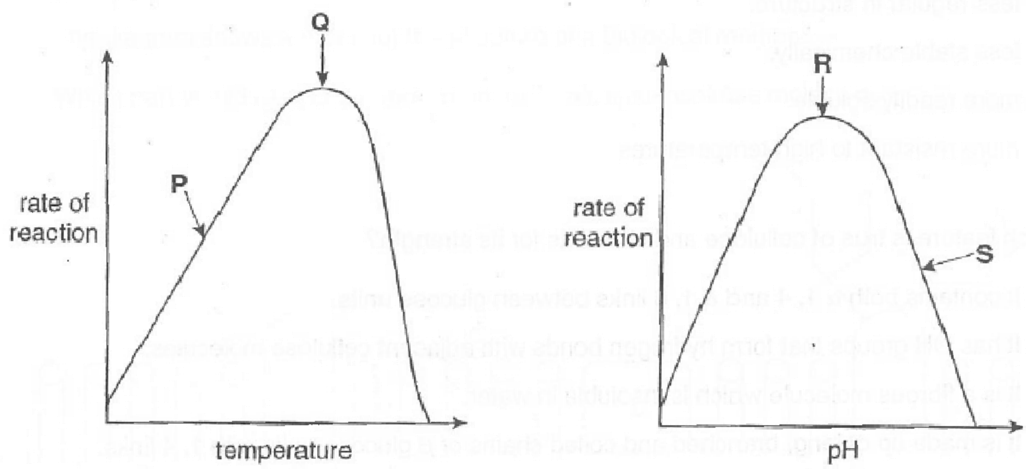
A Collagen has polypeptides arranged parallel to each other and the amino acid sequence contains a large variety of amino acids with different sized R-groups.

B Collagen has polypeptides that are arranged very closely together and the amino acid sequence has every third amino acid as glycine.

C Collagen has three polypeptides that are bounded to one another by covalent cross links and the amino acid sequence contains amino acids with hydrophobic R-groups.

D Collagen is an insoluble molecule and the amino acid sequence contains successive amino acids which are rotated to allow formation of bonds.

5 The graphs show the effects of temperature and pH on enzyme activity.



Which statement is a correct explanation of the rate of reaction at the point shown?

A At **P**, hydrogen bonds in the enzyme are broken.

B At **Q**, the kinetic energy of enzyme and substrate is highest.

C At **R**, covalent bonds are formed between enzyme and substrate.

D At **S**, ionic bonds in the enzyme are broken.

6 A cross between radish (*Raphanus sativus*) and cabbage (*Brassica oleracea*) produced the following results.

Radish *Raphanus sativus* (2n = 18)	X	Cabbage *Brassica oleracea* (2n = 18)

↓

Viable, infertile hybrid
(2n = 27)

↓

Brassicoraphanus
(2n = 54)

Which of the following statements are possible explanations for the results of the cross shown above?

1 Non-disjunction of all sister chromatids during anaphase II resulted in the production of diploid gametes in radish, which fused with a haploid gamete of cabbage to give rise to the infertile hybrid progeny.

2 The hybrid is infertile because it has a diploid number which is odd.

3 The hybrid is infertile as there are nine unpaired chromosomes during prophase I, which prevented the formation of gametes.

4 Addition of colchicine prevents the formation of spindle fibres, which enabled the infertile hybrid to form gametes. Fusion of two of these gametes gives rise to *Brassicoraphanus.*

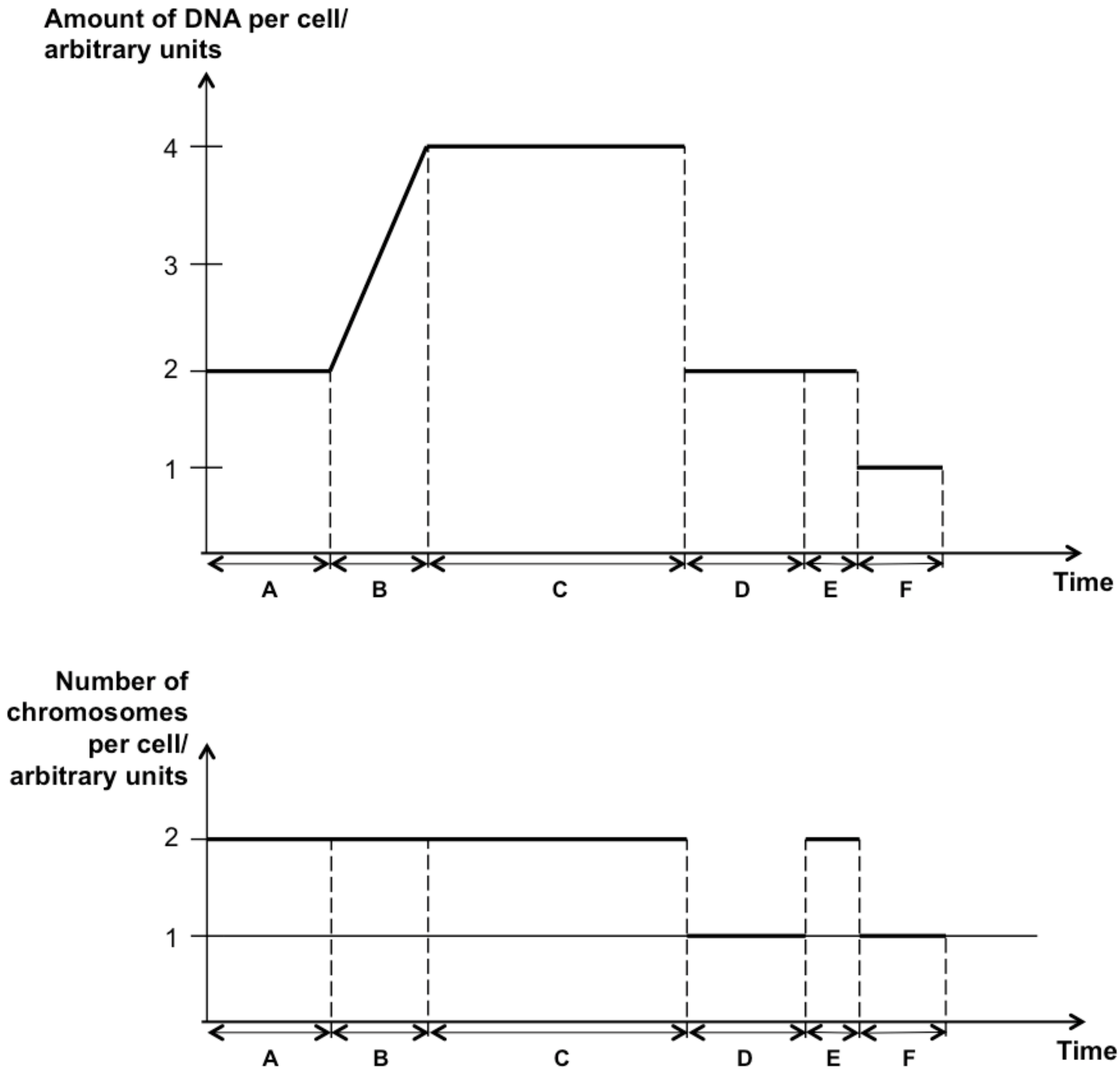
A 1 and 4

B 2 and 3

C 1, 2 and 4

D 1, 3 and 4

7 The figures below show how the amount of DNA and number of chromosomes vary in a cell of *Chrysanthemum makinoi* during meiosis.



Which of the following statement is **false**?

A Phase A and B correspond to G1 and S phase of interphase respectively.

B In phase C, the cell is undergoing prophase, metaphase, anaphase and telophase of meiosis I only.

C In phase E, the cell is undergoing telophase II only.

D In both phase D and F, the cell has completed cytokinesis.

8 The table below shows a list of characteristics displayed by mutant strains of *E. coli* during DNA replication and the possible reasons.

No.	**Characteristics**	**Enzymes or functions affected by mutation**
1	Okazaki fragments accumulate and DNA synthesis is never completed	DNA ligase activity is missing
2	Supercoils are found to remain at the regions that flank the replication bubbles	DNA helicase is hyperactive
3	Synthesis is very slow.	DNA polymerase keeps dissociating from the DNA and has to re-associate
4	No initiation of replication occurs.	The TATA box region at origin of replication is deleted

Which of the reasons correctly explain the characteristics displayed by the mutant *E. coli* strains?

A 1 and 3

B 2 and 3

C 1, 2 and 4

D 1, 3 and 4

9 Tay-Sachs disease is a fatal neurodegenerative disease which is caused by a mutation in the hexosaminidase A (Hex A) gene located on chromosome 15.

Part of the sequence of the non-template (coding) DNA strand of the normal Hex A allele and the mutated Tay-Sachs allele are shown below. The sequences are the same as the mRNA sequence of both alleles.

DNA sequences of normal Hex A allele:

Amino acid position		424	425	426	427	428	429	430	431	
Non-template DNA	5'…	CGT	ATA	TCC	TAT	GGC	CCT	GAC	TGT	…3'

DNA sequences of mutated Tay-Sachs allele:

Amino acid position		424	425	426	427	428	429	430	431	
Non-template DNA	5'…	CGT	ATA	TCT	ATC	CTA	TGG	CCC	TGA	…3'

For both alleles, 9 different amino acids are encoded for by the DNA triplets:

Amino acid	DNA triplet
Arg	CGT
Asp	GAC
Cys	TGG, TGT
Gly	GGC
Ile	ATA, ATC

Amino acid	DNA triplet
Leu	CTA
Pro	CCC, CCT
Ser	TCC, TCT
Tyr	TAT
Stop codon	TAG, TAA, TGA

Which statement is true?

A The disease is caused by the deletion of one DNA nucleotide.

B The Hex A protein encoded for by the Tay-Sachs allele is non-functional due to a frameshift mutation.

C The polypeptide encoded for by the Tay-Sachs allele has the same number of amino acids as that encoded by the normal Hex A allele.

D At amino acid position 431, there is a silent mutation.

10 Scientists were able to excise out the promoter of eukaryotic gene and attach it to the end of a gene as shown below. The gene was reintroduced back to the organism, which is homozygous at the gene loci.

Which of the following outcomes is true?

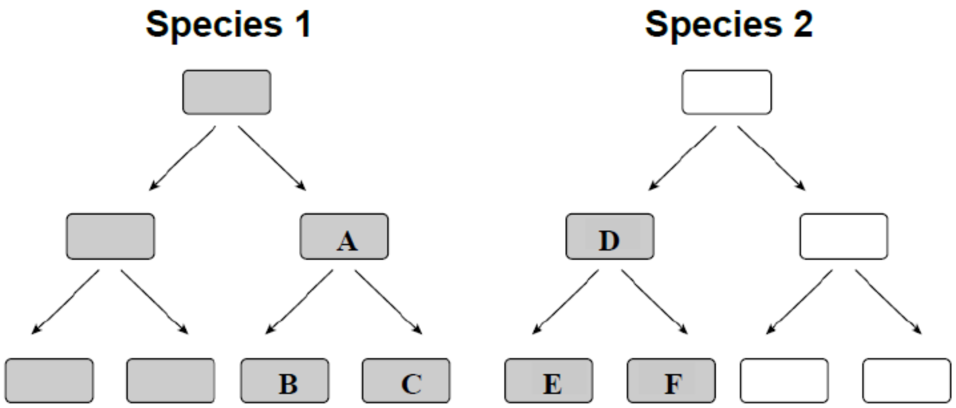
A No transcription of the gene takes place, as the promoter cannot initiate transcription of the non-template strand.

B Transcription initiates but may prematurely terminate due to a premature stop codon.

C Transcription takes place but translation cannot take place because the genetic code is no longer the original code.

D Transcription takes place but less translation occurs to obtain functional protein because the mRNA may form a duplex RNA.

11 The diagram below shows how two species of bacteria reproduce when placed together in a growth medium. The bacteria that are shaded are resistant to the antibiotic penicillin.



Which one of the following statement(s) is likely to be true?

1 Bacteria **B** and **C** are resistant to penicillin as a result of binary fission of Bacterium **A**.

2 Bacteria **C, D** and **F** are resistant to penicillin as a result of random mutation.

3 Bacterium **D** is resistant to penicillin as a result of conjugation from Bacterium **A**.

4 Bacterium **D** is resistant to penicillin through transduction from Bacterium **A** where there is transfer of the complete F plasmid.

A 3 only

B 1 and 3

C 1 and 4

D 2, 3 and 4

12 When a mutant strain of *Escherichia coli* that has lost the regulatory gene of its tryptophan operon is placed in a medium that contains all nutrients the cell need to grow except tryptophan, which of the following will occur?

A The cells will grow even though there is no tryptophan in the medium.

B The cells will grow until excessive tryptophan arrests the expression of the operon.

C The cells will not grow until enough tryptophan has been synthesised to make the repressor active.

D The cells will never grow unless tryptophan is added to the medium.

13 Temperate bacteriophages, such as the lambda phage, undergo the lysogenic cycle in their bacterial host cells.

Which of the following could prevent the temperate bacteriophages from entering the lysogenic life cycle?

1 A loss-of-function mutation in the viral gene which codes for integrase.

2 Deletion of 10 nucleotides at the site of phage integration on the bacterial chromosome.

3 Gain-of-function mutations in the viral genes which code for transcriptional repressor proteins.

4 Loss-of-function mutations in the viral genes which code for nucleases which break down bacterial chromosomal DNA.

A 1 and 2 only

B 1 and 3 only

C 2 and 4 only

D 1, 2 and 3 only

14 Which of the following occurs in the reproductive cycle of the human immunodeficiency virus (HIV) but **not** in that of the influenza virus?

A Newly synthesised viruses are released from host cells by budding from cell surface membrane.

B Host cell ribosomes are used to synthesise viral proteins.

C Viral RNA acts as a template for viral DNA synthesis.

D Viruses enter the host cell by endocytosis.

15 The percentage of the human genome that is transcribed is larger than predicted based on the range of proteins made by the cells.

Which of the following accounts for the difference?

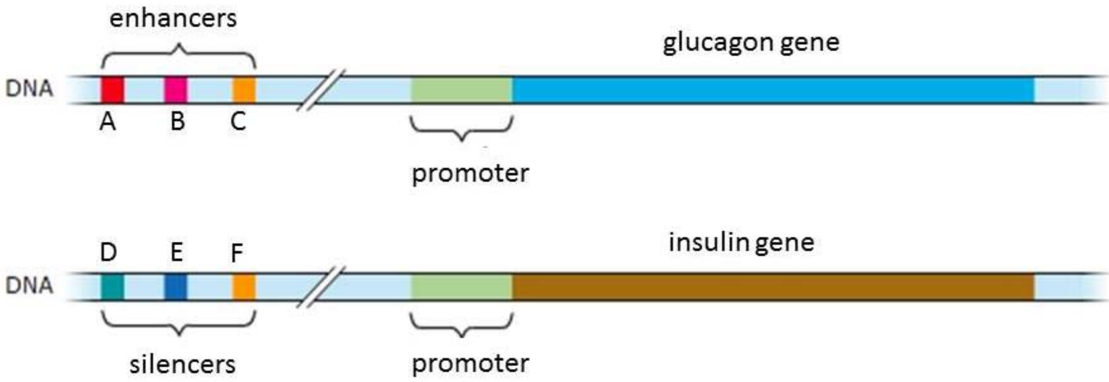
A Alternative splicing can result in more than one kind of protein produced from one gene.

B Some genes are transcribed to give RNA that is not meant to serve as a template for protein synthesis.

C The enhancers present in the human genome are also transcribed to bring about an increase in the transcription of protein-coding genes.

D The telomeric regions are also transcribed to give telomerase, which helps to maintain the telomere length.

16 The diagram below shows the control elements and two genes found in the human genome.



Which of the following statement(s) about the above genes is/are true?

1 The glucagon gene is found only in the α-cells of the Islets of Langerhans while the insulin gene is found only in the β-cells of the Islets of Langerhans.

2 Binding of control elements, specific transcription factors and RNA polymerase at the promoter initiates transcription of glucagon.

3 The glucagon gene will be transcribed at a high level when transcription factors bind to control elements A, B, and C.

4 The expression of insulin can only be suppressed when transcription factors bind to control elements D, E and F.

A 3 only

B 2 and 3 only

C 1, 2 and 4 only

D 2, 3 and 4 only

17 Which statement correctly describes introns and/or exons?

A Different combinations and numbers of exons and introns in the mature mRNA allow more than one type of protein to be coded for by a gene.

B Mutations that occur within introns have no effect on the primary structure of protein as it will not be present in the mature mRNA.

C The exons found in the DNA sequence of a gene are always translated into proteins.

D Mutation at a splice site would result in a truncated protein as the intron, which is not excised, is non-coding and cannot be translated.

18 Which of the following statements correctly describe the changes in cancer cells?

1. Limitless replicative potential often results in the accumulation of chromosomal mutations in many cancer cells.
2. Cancer cells could overproduce signal molecules so that they become self-sufficient in growth signals.
3. Angiogenesis is the result of expression of oncogenes in a cell line that produces blood vessels.
4. Loss-of-function mutations in tumour suppressor genes contribute to tissue invasion and metastasis.

A 1 and 4 only

B 2 and 3 only

C 1, 2 and 4 only

D 2, 3 and 4 only

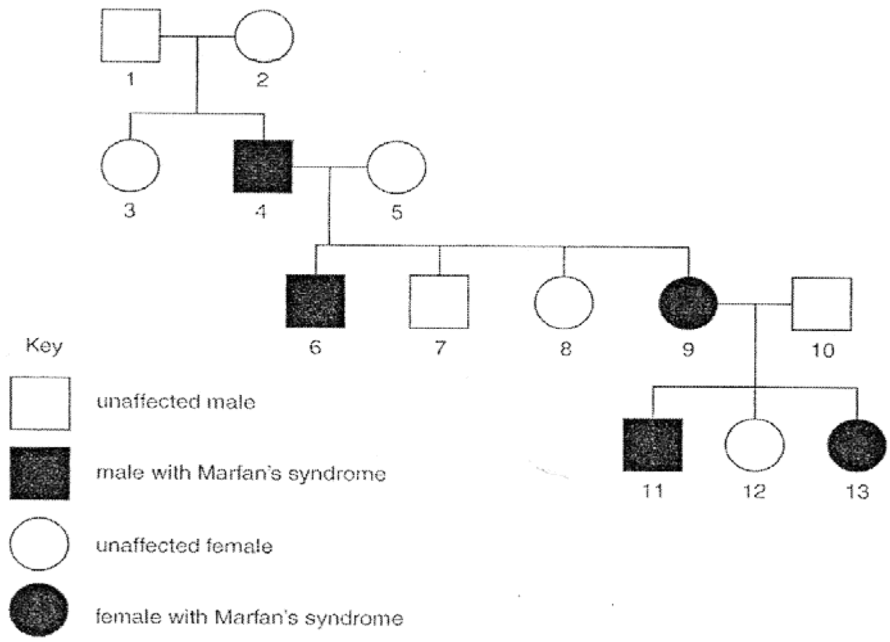
19 The statements listed below give information on the genetic control of hair curliness in dogs.

1. Chromosome 27 contains the gene responsible for curliness of the dog hair.
2. The nucleotide sequence of the gene produces an enzyme with arginine at residue position 151 but small changes in the nucleotide sequence produces an enzyme with cysteine at this point.
3. A dog may have both nucleotide sequences in its genome.
4. A dog producing both enzymes will have ‘wavy’ coat of hair.
5. At fertilisation, the dog inherits one set of chromosomes from each parent. Each set carries one of each form of the gene.

Which row matches each statement to the genetic term that it most closely describes?

	1	**2**	**3**	**4**	**5**
A	locus	genotype	allele	heterozygous	phenotype
B	locus	allele	heterozygous	phenotype	genotype
C	genotype	allele	heterozygous	phenotype	locus
D	locus	allele	genotype	phenotype	heterozygous

20 Marfan's syndrome is a rare genetic condition of humans, caused by a dominant allele. This condition is caused by a reduction in the quality or amount of the important protein fibrillin-1. The diagram below shows a family pedigree including some people with the condition.



Which of the following conclusions can be made based from the information provided?

A The fibrillin-1 gene locus is on the X chromosome.

B The mutation that gave rise to a non-functional fibrillin-1 allele occurred in the individual 4 during embryonic development.

C The variation in disease expression will be discontinuous.

D If individual 11 mated with a female who is heterozygous for the gene locus, the probability that they have an unaffected female child is 0.125.

21 In pigeons, the alleles of the gene controlling eye colour are co-dominant.

Two separate crosses were carried out and the results were shown below.

Cross 1

Parental generation	black eye female	X	white eye male
F_1 generation	grey eye females and black eye males		

Cross 2

Parental generation	white eye female	X	black eye male
F_1 generation	grey eye females and white eye males		

What phenotypic ratio would be expected in the F_2 generation in the first cross?

A 1 black-eyed male: 1 white-eyed male: 2 grey-eyed females

B 1 black-eyed male: 1 white-eyed male: 1 grey-eyed female: 1 black-eyed female

C 1 black-eyed male: 1 grey-eyed male: 1 white-eyed female: 1 black-eyed female

D 2 black-eyed males: 1 grey-eyed female: 1 white-eyed female

22 The coat colour of Labrador retrievers is controlled by two genes, **B/b** and **A/a**. Allele **B** (dominant) codes for black coat, while allele **b** (recessive) codes for brown coat. The coat colour of a Labrador retriever with genotype **aa** is yellow.

A cross between a male black Labrador retriever and a female yellow Labrador retriever produced some black puppies and some yellow ones.

What are the genotypes of the parental dogs?

	Black retriever	**Yellow retriever**
A	AaBb	aabb
B	AaBb	aaBb
C	AaBb	aaBB
D	AABb	aaBb

23 A cross was made between 2 pure breeding maize varieties, Tom Thumb and Black Mexican which differed markedly in ear length. The ear length for the parental, F_1 and F_2 generations was measured in centimetres and recorded in the table below with the number of ears in each length category (e.g.13 of the F_1 plants produced ears 12 cm in length).

Generation	Ear length (cm)														
	5	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**13**	**14**	**15**	**16**	**17**	**18**	**19**
Black Mexican (parental)									4	11	12	32	27	10	8
Tom Thumb (parental)	4	28	19	3											
F_1					2	12	18	13	17						
F_2			2	3	15	22	33	28	19	13	2	2			

Which of the following statements correctly explains the data shown above?

1 The phenotypic variation is continuous and could be the result of multiple alleles.

2 Variation in environmental factors has a large effect on the phenotype.

3 The increase in phenotypic variation from F_1 to F_2 generation could be due to genetic mutations.

4 The increase in phenotypic variation from F_1 to F_2 generation could be due to sexual reproduction processes like crossing over, independent assortment of homologous chromosomes and random fusion of gametes.

A 1 and 3

B 2 and 4

C 1, 2 and 4

D 2, 3 and 4

24 Unaffected carriers with chromosomal inversions are likely to produce genetically abnormal progeny.

Which of the following explains this?

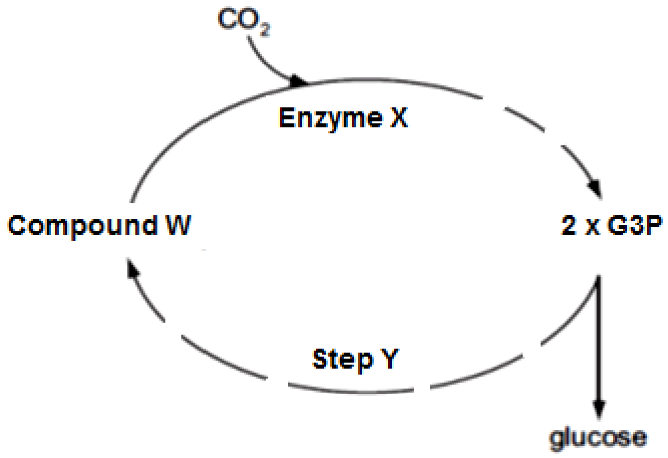
A The mutated chromosome is more likely to be placed in a gamete than the normal chromosome.

B The mutated chromosome is unable to accomplish synapsis with the normal chromosome during meiosis.

C Crossovers cannot occur between normal and mutated chromosomes.

D Crossovers between the normal and mutated chromosomes lead to chromosomes with deletions, deficiencies, or abnormal structure.

25 The figure below summaries some key reactions which occur in the Calvin cycle. The dashed lines indicate that there is more than one reaction present.



Using the figure above and your knowledge of Calvin cycle, determine which one of the following statements below is true.

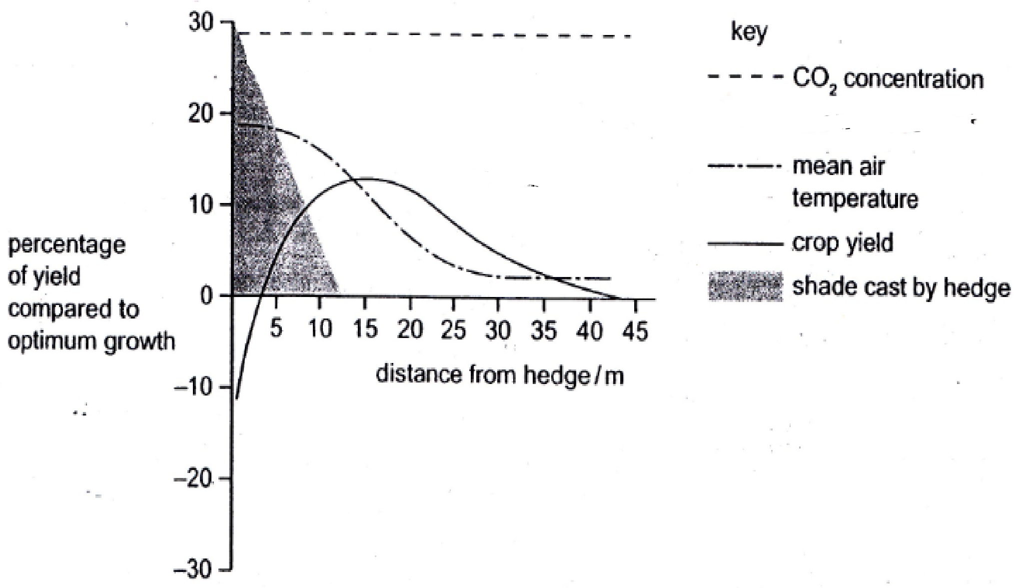
A Compound **W** is expected to accumulate if carbon dioxide concentration increases under low light intensity.

B Products released from Enzyme **X** are expected to accumulate when light intensity is low.

C G3P is expected to accumulate when light intensity is low.

D ATP from substrate level phosphorylation is required for Step **Y** to proceed and Compound **W** to be formed.

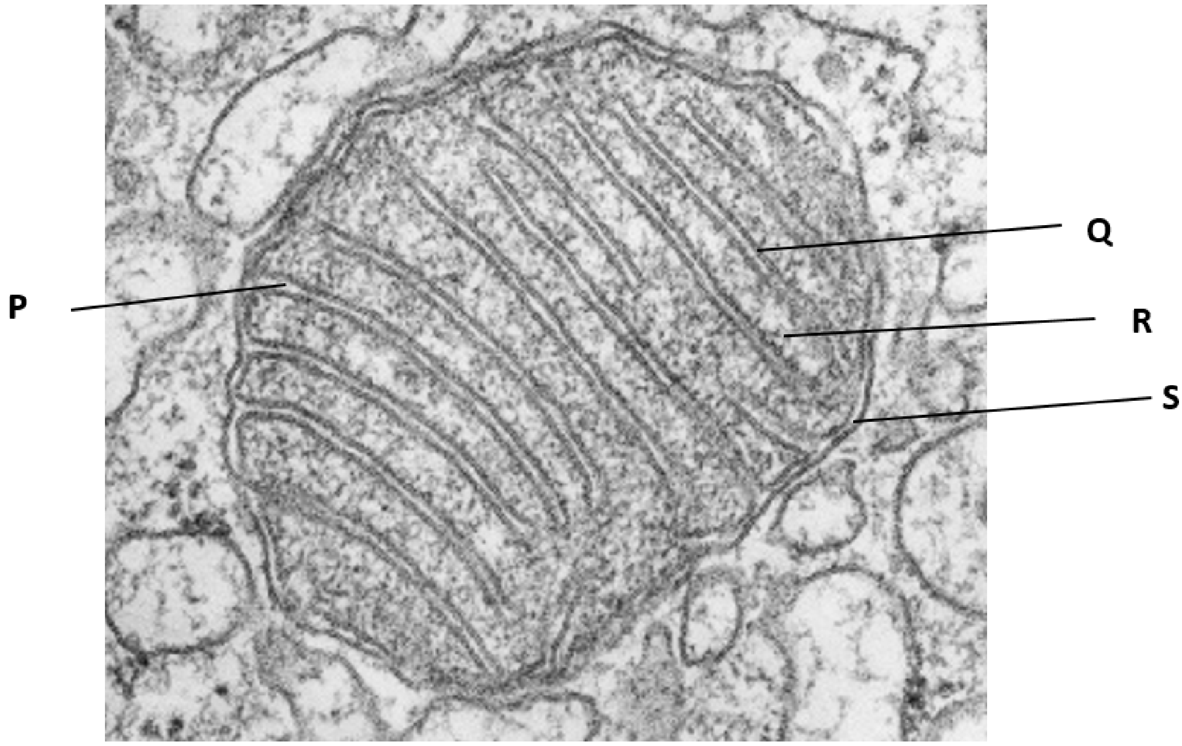
26 In the diagram, researchers superimposed the changes in four environmental factors on a graph of crop yield against distance from a hedge.



Which factor is most likely to have been the limiting factor in crops 8m and 25m from the hedge?

	Limiting Factor	
	8m from hedge	**25m from hedge**
A	CO_2 concentration	Light intensity
B	Light intensity	Mean air temperature
C	Mean air temperature	Mean air temperature
D	Light intensity	CO_2 concentration

27 The figure below shows an electron micrograph of a mitochondrion.



Match the following processes with each of the labelled sites **P – S.**

1 Oxidative decarboxylation
2 Lowering of pH
3 Protein synthesis
4 Electron flow
5 Formation of oxidised co-enzymes of dehydrogenase

	P	Q	R	S
A	1, 3, 5	2	4, 5	3
B	2	4, 5	1, 3	2
C	1, 5	3, 4	2	1
D	2	4	1, 3, 5	2

28 Two test tubes containing the following contents are shown below:

Tube 1:
Radioactive glucose solution + yeast cells suspension + oxygen + antimycin

Tube 2:
Radioactive glucose solution + yeast cells suspension + oxygen

Radioactive glucose has all its six carbons made of radioactive ^{14}C. The initial radioactivity measured for the glucose in each test tube is 60 arbitrary units.

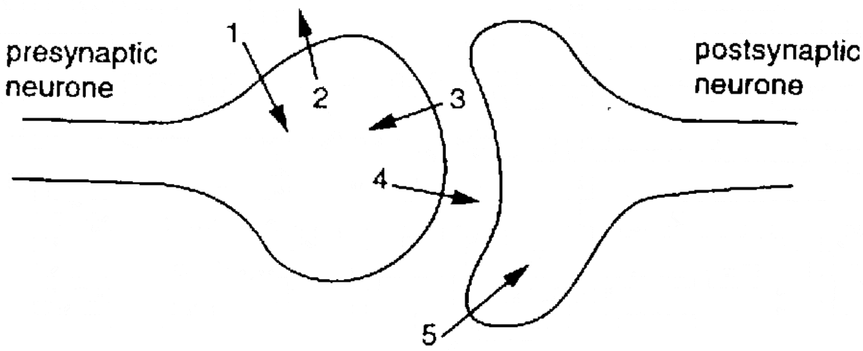
Antimycin is an electron transport chain inhibitor.

If the gaseous product and the aqueous products are tested using a radioactive meter after all the glucose has been metabolised, what would be the final observed readings?

	Tube 1 *(radioactivity measured/ arbitrary units)*		**Tube 2** *(radioactivity measured/ arbitrary units)*	
	aqueous products	**gaseous products**	**aqueous products**	**gaseous products**
A	0	60	40	20
B	20	40	0	60
C	40	20	0	60
D	40	20	60	0

29 The diagram shows the sequence of events occurring as an action potential arrives at a synapse.

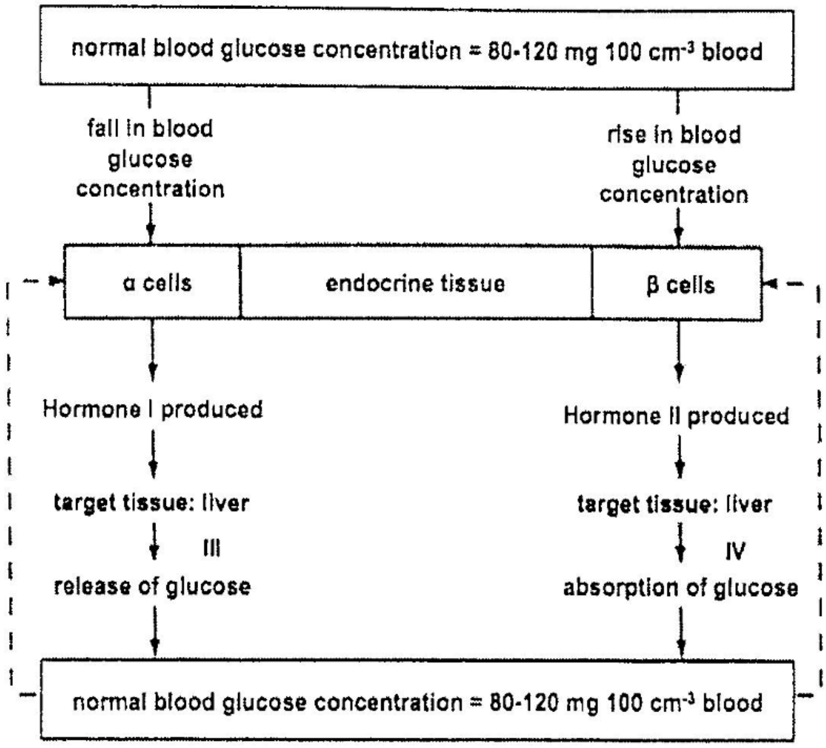
The numbered arrows represent the movement of substance across the membranes.



What are the substances moving across the membranes?

	1	**2**	**3**	**4**	**5**
A	K^+	Na^+	acetylcholine	Ca^{2+}	K^+
B	K^+	Na^+	K^+	Ca^{2+}	acetylcholine
C	Na^+	K^+	Ca^{2+}	acetylcholine	Na^+
D	Na^+	K^+	Na^+	acetylcholine	Ca^{2+}

30 The diagram below shows the role of an endocrine tissue in controlling blood glucose concentration.



Which of the following statements is/are true?

1 Small amounts of hormone I and II inducing a large response in liver tissue demonstrates positive feedback.

2 Hormones I and II inducing different responses from the same target tissue is due to the hormones binding to different receptors on the liver cell surface membrane.

3 The binding of Hormone I to receptors on liver cell surface membrane leads to the activation of second messenger cAMP.

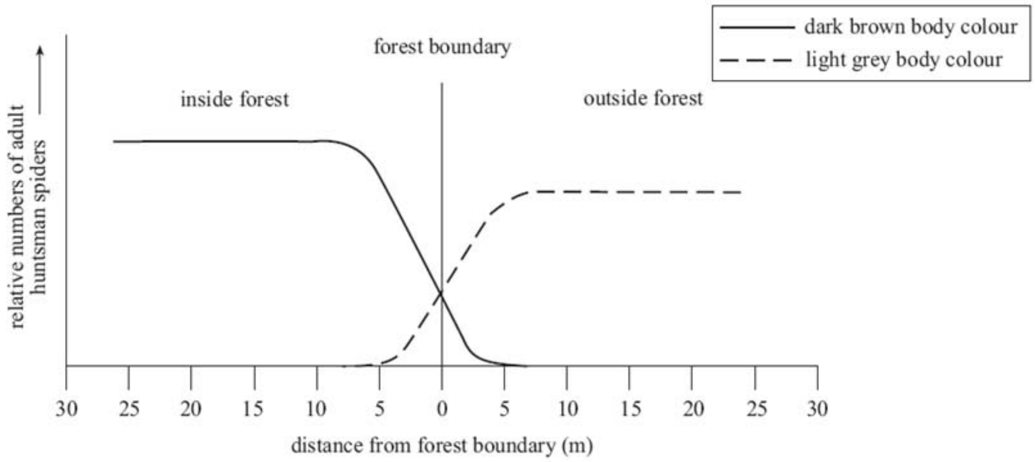
A 1 only

B 2 only

C 1 and 3

D 2 and 3

31 The graph below shows the distribution of huntsman spiders at a forest boundary:



One species of huntsman spider (*Isopeda isopedella*) varies in body colour from dark brown to light grey. In one community at the forest boundary, two populations of this species were found. Some were found living inside the forest and others were found living just outside the forest. The relative numbers of dark brown adult spiders and light grey adult spiders found at certain distances from the forest boundary are shown in the graph above.

Which of the following can be **least** inferred from the graph?

A The type of selective advantage inside the forest is different from the type of selective advantage outside the forest.

B The plateau in the population number as seen inside the forest and outside the forest is due to competition among the adult spiders for limited resources.

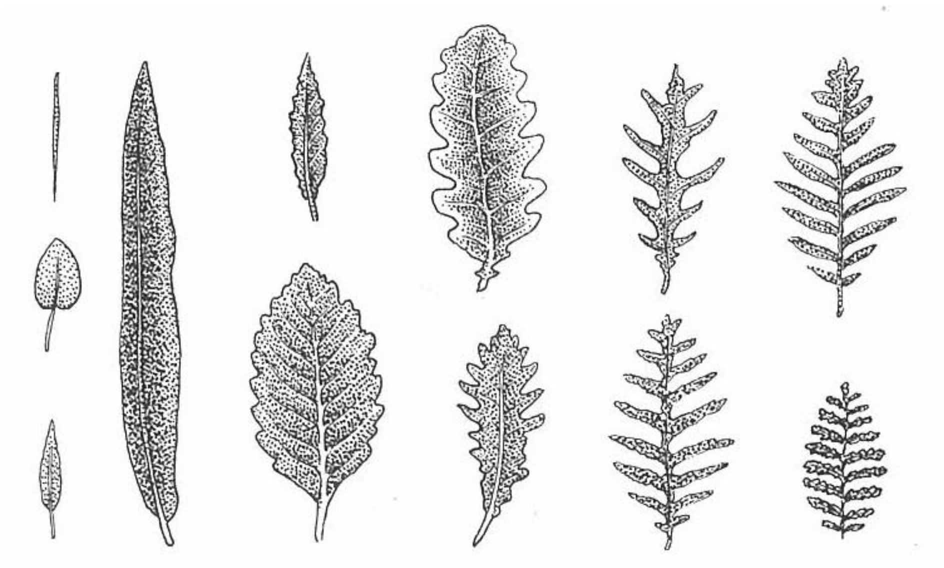
C The lower plateau of spider population outside the forest compared to that of inside the forest is due to the presence of additional selection pressure existing outside the forest.

D Dark brown huntsman spiders are not eaten by birds inside the forest as their colour allows them to camouflage and hence provides a selective advantage.

32 One of the six native genera of *Lobeliaceae, Cyanea,* contains 55 species that constitute 6% of the flora in Hawaii. They are restricted to particular islands or parts of islands. All *Cyanea* descend from a single ancestor, many have undergone striking changes in growth form, leaf size and shape.

Leaf morphology in *Cyanea* is thought to be determined by several gene loci on separate chromosomes.

The leaf morphologies of different *Cyanea* species are shown below.



Which of the following reasons explain the emergence of distinct leaf morphological differences between various species?

1 Mutation in leaf cells

2 Crossing over between non-sister chromatids of homologous chromosomes during gamete formation

3 Different humidity levels on different islands and in different parts of an island

4 The Hawaiian honeycreepers which pollinate the flowers of the plants on one island are unable to fly to other islands

A 1 and 2 only

B 3 and 4 only

C 1, 3 and 4 only

D 2, 3 and 4 only

33 Which of the following statements about the Neutral Theory of Molecular Evolution is **incorrect**?

A The neutral theory involves studying the changes in neutral allele frequencies of a gene pool in the absence of selection.

B The neutral theory involves nucleotide sequence variations that do not affect the reproductive success of organisms.

C The neutral theory contradicts Darwin's theory of natural selection as it postulates that change in allele frequency is driven by genetic drift and not natural selection.

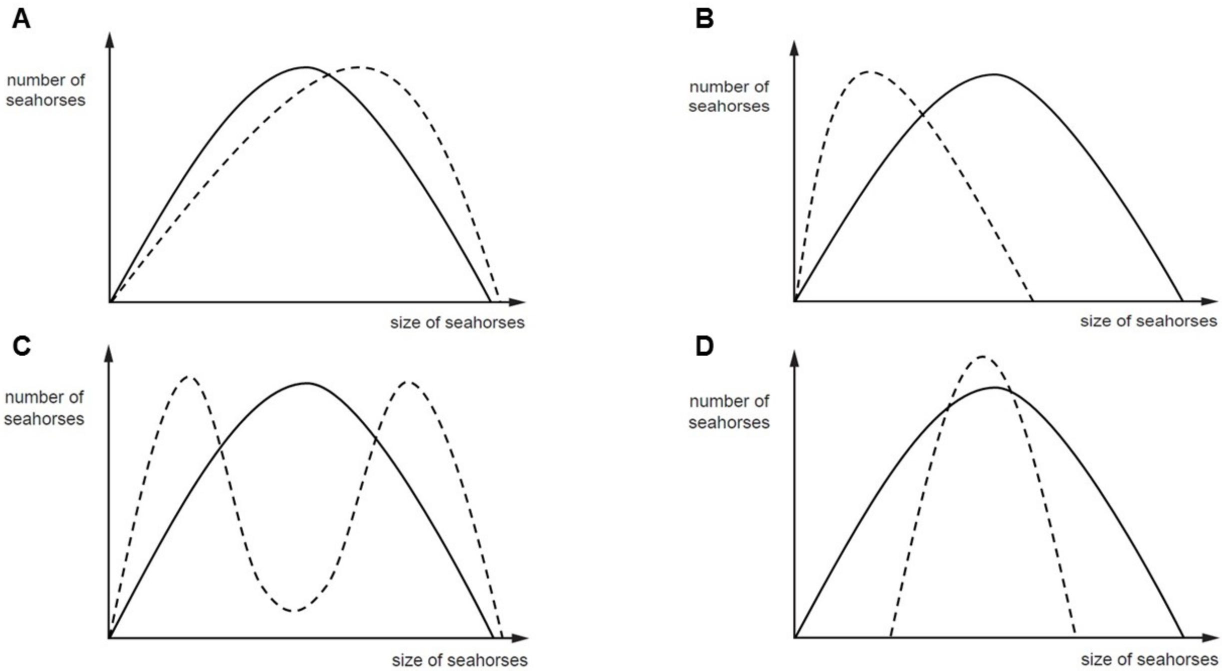
D Rate of change in gene sequence may be greater than the rate of change in amino acid sequence of the polypeptide chain coded for due to degeneracy of the genetic code.

34 The seahorse, *Hippocampus,* is an unusual small fish. It gives birth to live young and it is the male rather than the female that becomes pregnant.

In one species of seahorse, large females within a population mate with large males and small females mate small males. Few medium-sized individuals are produced and they have a low survival rate.

Which graph shows the effect of natural selection on size of seahorses after a fixed period of time?

Legend: ——— Original population – – – Population after selection



35 Which of the following statements best explains why some restriction sites are suitable to be used in genetic engineering?

A A short nucleotide sequence that can be recognised by many types of endonucleases.

B A short nucleotide sequence that occurs many times in a DNA molecule such that cutting the DNA molecule with an endonuclease will result in many fragments.

C A short nucleotide sequence that can be cut by an endonuclease to produce single-stranded ends that can bind to other single-stranded ends.

D A short nucleotide sequence that allow a fragment of DNA to be introduced into another DNA molecule via homologous recombination.

36 Which of the following factors is **not** critical for the functioning of Polymerase Chain Reaction (PCR)?

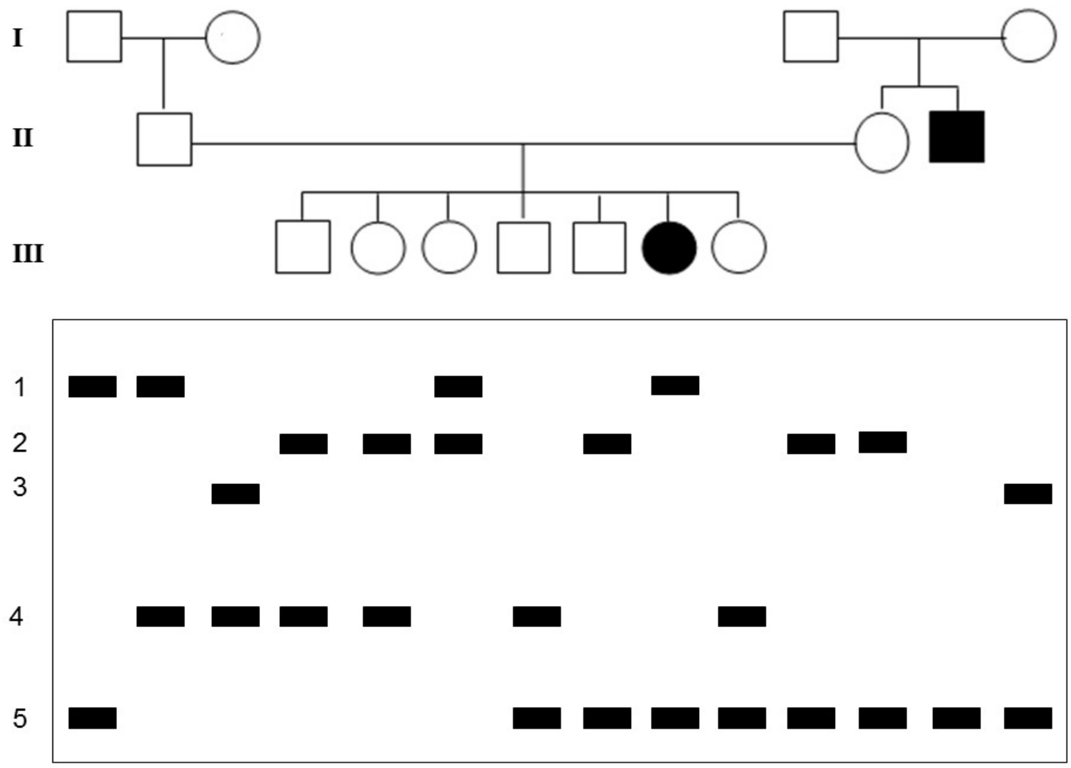
A Presence of DNA sequences other than target DNA.

B Temperatures for Stage 1 and 3 of the PCR cycle are only sustained for a short period of time (1 – 5 minutes).

C Presence of large amount of primers such that it is not the limiting reagent.

D Use of a thermostable polymerase enzyme.

37 A Restriction Fragment Length Polymorphism (RFLP) locus with 5 alleles (1-5), is linked to the gene causing Disease H.



Which of the following conclusions can be made from the information given above?

1. Crossing over occurred at a locus between the gene locus and the RFLP locus in the affected individual from generation III.
2. The disease is an autosomal recessive disease.
3. Among the unaffected individuals, all are heterozygous at the gene locus.
4. The RFLP alleles have different number of tandem repeats between 2 restriction sites.

A 1 and 3

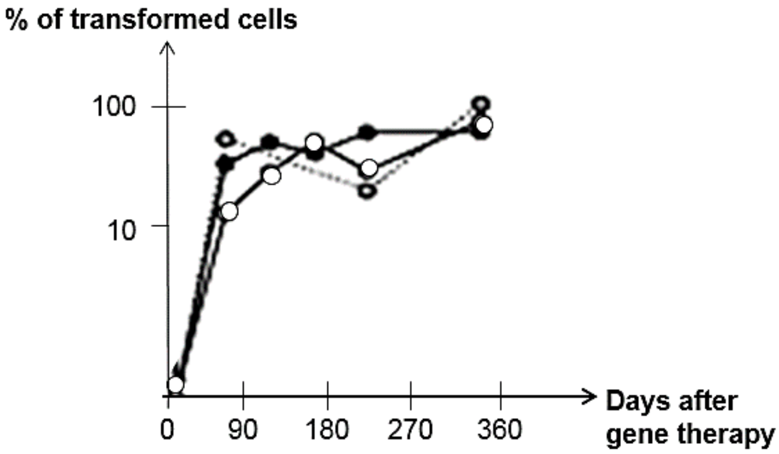
B 2 and 4

C 1, 2 and 4

D 2, 3 and 4

38 Recent research has developed a technique that combines gene therapy with stem cell therapy. This inspired a group of scientists to carry out a clinical trial with such modified stem cells in the hope of treating a neurological disease that is caused by a homozygous recessive genotype at a single gene locus.

Upon introducing stem cells with the functional dominant allele into the brains of three patients, their conditions did not improve although the following results were obtained from analysing the cell count in the target area of their brains.



What would be the most appropriate investigation to carry out in view of the unsuccessful clinical trial?

- **A** regulation of neurone-specific genes in implanted stem cells
- **B** in-vitro efficiency of gene delivery vector
- **C** modification of stem cells to reduce immune reaction to implanted stem cell
- **D** plasticity of neural stem cells to form other types of specialised cells

39 Plant physiologists attempted to produce papaya plants using tissue culture. They investigated the effects of different concentrations of plant growth factors on small pieces of the stem tip from a papaya plant. Their results are shown in the table below.

Concentration of auxin / $\mu mol\ dm^{-3}$	Concentration of cytokinin / $\mu mol\ dm^{-3}$		
	5	25	50
0	No effect	No effect	Leaves produced
1	No effect	Leaves produced	Leaves produced
5	No effect	Leaves produced	Leaves and some plantlets produced
10	Callus produced	Leaves and some plantlets produced	Plantlets produced
15	Callus produced	Callus and some leaves produced	Callus and some leaves produced

What evidence from the table supports that cells from the stem tip are totipotent?

A Under certain concentrations of cytokinin to auxin, there is production of both leaves and calluses or leaves and plantlets showing that more than one type of cells can be produced at any one time.

B Stem tip cells are able to produce calluses when the concentration of cytokinin to auxin is in the ratio of 1:2 or 1:3.

C Stem tip cells are able to produce leaves although they are supposed to differentiate into stems.

D Stem tip cells are able to give rise to plantlets.

40 What is an example of genetically modified organisms?

A Cows that grow to adult size quickly due to injection of growth hormone into the cows.

B Pigs that grow to large size via inbreeding.

C High yielding rice that are flood resistant due to intensive self-pollination over many generations.

D Durians that ripen slowly due to anti-sense RNA technology.

Paper 1 Answer Scheme

Qns	Ans
1	C
2	B
3	A
4	B
5	D
6	D
7	C
8	A
9	B
10	D
11	B
12	A
13	A
14	C
15	B
16	A
17	B
18	C
19	B
20	D

Qns	Ans
21	B
22	C
23	B
24	D
25	B
26	B
27	B
28	C
29	C
30	D
31	D
32	B
33	C
34	C
35	C
36	A
37	B
38	A
39	D
40	D